2013 IEEE International Conference on Robotics and Automation (ICRA)
Karlsruhe, Germany, May 6-10, 2013

Position Control of Concentric-Tube Continuum Robots using a Modified Jacobian-Based Approach

Ran Xu, Ali Asadian, Anish S. Naidu, and Rajni V. Patel

***Abstract*— Concentric-tube robots can offer dexterous positioning even in a small constrained environment. This technology turns out to be beneficial in many classes of minimally invasive procedures. However, one of the barriers to the practical use of a concentric-tube robot is the design of a real-time control scheme. In previous work by the authors, a computationally efficient torsionally compliant kinematic model of a concentric-tube robot was developed. Using this computationally fast technique and deriving the robot's Jacobian, a new position control approach is proposed in this paper. This mechanism provides computational efficiency as well as good tracking accuracy. To evaluate the performance, experiments were conducted, and the results obtained demonstrate the feasibility of enabling the robot's tip to perform trajectory tracking in real time.**

I. INTRODUCTION AND PRIOR WORK

To minimize collateral damage to live tissue and access confined areas in minimally invasive procedures, surgical interventions are required that follow complex curved paths inside soft tissue. To this end, flexible needles have been widely used [1] as primary tools while erroneous guidance reduces the effectiveness of the planned therapy or diagnosis.

As an alternative, a concentric-tube robot as a subset of continuum robots is a new technology which provides more dexterity compared to a standard bevel-tip flexible needle or a catheter while having almost the same size [2]–[10]. This category of instruments has the potential to facilitate targeting in applications in which a complex 3D curvature is required. Another advantage of this developing family of tools is safety enhancement due to its inherent compliance compared with the traditional rigid surgical tools; so they can offer a suitable compromise between stiffness and curvature control. Furthermore, the shaft of the tubes can accommodate cables for controlling articulated tools mounted at the tip. For more details on design and analysis, please see [2], [3].

Note that compared with traditional robotic arms, this family of robots lacks rigid links and discrete joints. Thus, their kinematics cannot be represented solely in terms of constrained motion between rigid bodies, and it must include the deformation of the individual tubes as well. This fact adds more complexity to the modeling and therefore to the real-time control problem for this kind of robotic structures. In this regard, various modeling schemes for concentric-tube robots have been developed over the last few years; however, a variety of mechanical phenomena, e.g., torsion, nonlinear elasticity, and friction, have been ignored in order to simplify the modeling steps or accelerate the process time.

Under certain assumptions on the geometry of the tubes, closed-form forward kinematics can be represented by means of algebraic expressions. Finding the inverse kinematics is also not straightforward in general due to the nonlinear mapping between relative tube displacements and tip configuration as well as due to the multiplicity of solutions. In [4], the tubes were approximated as rigid in torsion and frictionless with piecewise constant curvatures, and forward and inverse kinematic equations were derived. Sears and Dupont presented a generalized inverse Jacobian method for solving the inverse kinematics disregarding the torsion [5].

The importance of including torsional effects was shown in [2], [3], [6], [7]. Dupont et al. [2] developed a torsion model that was applicable to any number of tubes whose stiffness and initial curvature could be arbitrary functions of arc length. In [3], a kinematic model containing only the torsion component of the straight segments was introduced. However, the authors hypothesized that torsion in the curved sections was the most significant unmodeled effect. Webster et al. [6], derived the differential kinematics of a general n-tube active cannula while accounting for torsional compliance in order to improve tip pose prediction. A compliant model including the torsional effects in both straight and curved sections significantly improved accuracy in [7]; however, this model was computationally very expensive.

Deformation of a concentric-tube robot in response to the external contact forces is another important issue. Lock et al. [8] showed that compared with the unloaded model, tip loading could increase the mean tip error by almost 50%. Hence, they developed a quasi-static model relating the externally applied loads to the robot's shape and tip configuration. Rucker et al. [9] applied Cosserat-rod theory to model forward kinematics to describe large deflections as a result of external point or distributed wrench loads.

Towards position control as the main motivation of our study, Dupont et al. [2] developed a functional approximation method to include torsional compliance and achieve computational efficiency for position control. However, this

Ran Xu, Ali Asadian, and Rajni V. Patel are with Canadian Surgical Technologies and Advanced Robotics (CSTAR), Lawson Health Research Institute, London, ON N6A 5A5, Canada, and with the Department of Electrical and Computer Engineering, Western University (The University of Western Ontario), London, ON N6A 5B9, Canada. Anish S. Naidu is with CSTAR and the Department of Mechanical and Materials Engineering, Western University. R.V. Patel is also with the Department of Surgery, Western University (emails: rxu25@uwo.ca, aasadian@uwo.ca, asriniva@uwo.ca, rvpatel@uwo.ca).

This work was supported by the Natural Sciences and Engineering Research Council (NSERC) of Canada under the grant RGPIN1345 (R.V. Patel). Partial financial support for R. Xu was provided by a postgraduate scholarship from the Peoples Republic of China. Financial support for A. Asadian was provided by an NSERC Collaborative Research and Training Experience (CREATE) program grant #371322-2009 in Computer-Assisted Medical Interventions (CAMI).

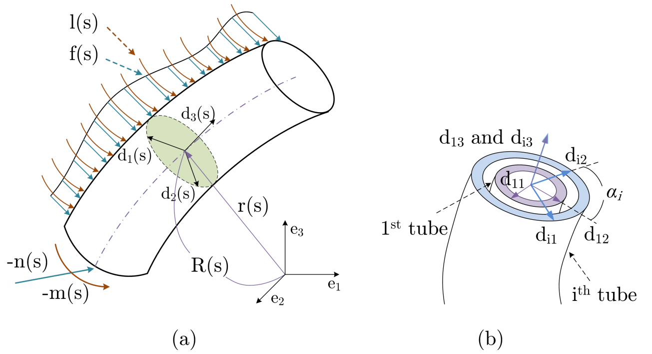


Fig. 1. (a) Cosserat beam (b) cross section of a concentric-tube robot showing coordinate frames associated with the 1st and the ith tubes.

approach required pre-computation of the forward kinematics over the entire workspace and then its approximation using a truncated Fourier series. The research described in this paper is in line with our previous work [10] and focuses on developing a feasible control strategy for concentric-tube robots without sacrificing accuracy. As is well known, computational cost places limitations on real-time implementations. To overcome this problem, we introduced a fast torsionally compliant kinematic model for a concentric-tube robot using Cosserat theory [10]. In the current study, we exploit this model to control the position of the robot's tip using the robot's Jacobian while the torsional effects for both straight and curved sections are incorporated.

The rest of this paper is organized as follows. Section II reviews our Cosserat-rod-based model, and introduces the resultant Jacobian. The mechanism utilized for position control is also described in this section. Section III presents a simulation study and this is followed by an experimental evaluation in section IV. Finally, section V concludes the paper with suggestions for future work.

II. A MODIFIED JACOBIAN-BASED STRATEGY FOR POSITION CONTROL

A. *A Fast Torsionally Compliant Model*

Following the procedure outlined in [10], we developed a fast model which included torsional effects along the entire robotic arm. This framework is summarized here to provide the basis for the rest of the study.

Concentric-tube robots can be considered as a set of elastic rods undergoing distributed force and torque generated by mechanical interactions. According to the Cosserat theory, each single elastic rod satisfies the equilibrium equations

$$\dot{n}(s) + u(s) \times n(s) + f(s) = 0 \tag{1}$$

$$\dot{m}(s) + u(s) \times m(s) + v(s) \times n(s) + l(s) = 0 \tag{2}$$

where $m(s)$ and $n(s)$ are the moment and stress vectors in the cross-section in terms of the length variable s. $f(s)$ and $l(s)$ are distributed force and torque along the rod, respectively. Herein, vector $u(s)$ consists of bending and torsional curvatures, while $v(s)$ includes shear deformation and elongation. Accordingly, the kinematics of an elastic rod can be completely defined by the vectors u and v expressed in the body frame $\{d_1(s), d_2(s), d_3(s)\}$ shown in Fig. 1(a).

Equations (1) and (2) can be extended to multiple elastic rods [2], and forwards kinematics referring to the mapping between joint space and Cartesian space variables can be represented by the following equations in which the inputs and outputs are both expressed in Cartesian space [10]. In order to improve computational efficiency, the kinematics is reformulated using Euler angles as shown below.

$$\begin{aligned} \dot{\theta}_1(s) &= \frac{u_{ix}(s)cos\theta_{i3}(s) - u_{iy}(s)sin\theta_{i3}(s)}{cos\theta_2(s)} \\ \dot{\theta}_2(s) &= u_{ix}(s)sin\theta_{i3}(s) + u_{iy}(s)cos\theta_{i3}(s) \\ \dot{\theta}_{i3}(s) &= u_{iz}(s) - \dot{\theta}_1(s)sin\theta_2(s) \end{aligned} \tag{3}$$

$$\begin{aligned} \dot{x}(s) &= sin\theta_2(s) \\ \dot{y}(s) &= -sin\theta_1(s)cos\theta_2(s) \\ \dot{z}(s) &= cos\theta_1(s)cos\theta_2(s) \end{aligned} \tag{4}$$

$$\begin{aligned} \begin{bmatrix} u_{ix}(s) \\ u_{iy}(s) \end{bmatrix} = \Big(\big(\sum_{j=1}^{n} K_j\big)^{-1} R_Z^T(\theta_{j3}(s) - \theta_{13}(s)) \\ \times \sum_{j=1}^{n} R_Z^T(\theta_{j3}(s) - \theta_{13}(s)) K_j \begin{bmatrix} \hat{u}_{jx}(s) \\ \hat{u}_{jy}(s) \end{bmatrix} \Big) \end{aligned} \tag{5}$$

$$\dot{u}_{iz}(s) = \frac{k_{ix}(s)}{k_{iz}(s)} \big(u_{ix}(s)\hat{u}_{iy}(s) - u_{iy}(s)\hat{u}_{ix}(s)\big) \tag{6}$$

in which, the subscript i refers to the tube number. $\hat{u}_i$ is also the initial curvature value while u_i represents its value after conformation. Here, $K_i = diag(k_{ix}, k_{iy}, k_{iz})$ denotes the stiffness matrix calculated from Young's modulus and moment of inertia, and θ_{i3} is the 3rd Euler angle of the ith tube. The transformation between the body frames of the 1st and the ith tubes is also represented by a pure rotation denoted by the matrix R_Z.

B. *Jacobian Derivation for the Concentric-Tube Robots*

Solving equations (3)-(6), the robot's position and orientation are directly obtained with no need to perform extra calculation to convert local curvatures into Cartesian coordinates. However, the reformulated structure includes a nonlinear differential equation; so linearization is employed to derive the closed-form solution. To this end, let us rearrange the above kinematic model in a compact form in which $\Theta(s) = [\theta_1(s) \;\; \theta_2(s) \;\; \theta_{i3}(s)]^T$ and $X(s) = [x(s) \; y(s) \; z(s)]^T$. Applying the Taylor series expansion and keeping the first-order terms, we have

$$\begin{bmatrix} \dot{u}_{iz}(s) \\ \dot{\Theta}(s) \\ \dot{X}(s) \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \\ g_{31} & g_{32} \end{bmatrix} \begin{bmatrix} 1 \\ s \end{bmatrix} \tag{7}$$

in which, $g_{ij} = g_{ij}(u_{iz}(0), \Theta(0), X(0))$. Thus, the sought solution is produced by integration of (7) with respect to s.

$$F = \begin{bmatrix} u_{iz}(s) \\ \Theta(s) \\ X(s) \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \\ g_{31} & g_{32} \end{bmatrix} \begin{bmatrix} s \\ \frac{1}{2}s^2 \end{bmatrix} + \begin{bmatrix} u_{iz}(0) \\ \Theta(0) \\ X(0) \end{bmatrix} \tag{8}$$

Due to linearization, the function approximation in (8) is accurate to within small intervals so this numerical

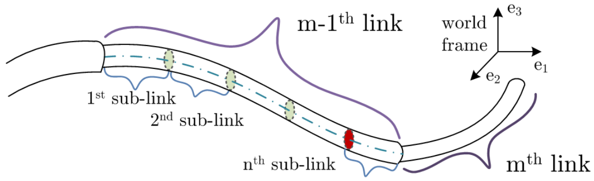


Fig. 2. Links and sub-links of a concentric-tube robot.

technique can be considered as a piecewise closed-form solution. Hence, prior to its use, the entire robotic arm needs to be divided into links and small sub-links as shown in Fig. 2. Successive approximation for each sub-link is then performed. As a result, forward kinematics of the (m-1)th link is obtained as $\tilde{F}^{m-1} = F^n(F^{n-1}(...F^1(u^1_{iz}(0), \Theta^1(0), X^1(0), l^1)..., l^{n-1}), l^n)$. Herein, for the jth sub-link $(1 \leq j \leq n)$, l^j is the corresponding length while the function F^j maps the vector $[u^j_{iz}(s)\ \Theta^j(s)\ X^j(s)]^T$ evaluated at its proximal end to the same vector evaluated at the sub-link's distal end using (8). Moreover, $l^j = H^j(p_1, p_2, ..., p_k)$ where k is the number of tubes in the entire robot, and p is the position of the proximal end. Finally, the forward kinematics of the entire robot is consecutively obtained in a similar manner, i.e., $\tilde{F} = \tilde{F}^m(\tilde{F}^{m-1}(...\tilde{F}^1()...))$.

In conjunction with the analytical solution introduced in section II-A, the closed-form forward kinematics is differentiated to produce the Jacobian matrix for each sub-link. It will be employed later on for control of concentric-tube robots. Using (8), the Jacobian is derived as

$$J^j_G = \begin{bmatrix} J^j_{g1} & J^j_{g2} \\ 0 & I_{k\times k} \end{bmatrix} \tag{9}$$

where

$$J^j_{g1} = \begin{bmatrix} \frac{\partial F^j}{\partial u^j_{iz}(0)} & \frac{\partial F^j}{\partial \Theta^j(0)} & \frac{\partial F^j}{\partial X^j(0)} \end{bmatrix} \tag{10}$$

$$J^j_{g2} = \begin{bmatrix} \frac{\partial F^j}{\partial H^j}\frac{\partial H^j}{\partial p_1} & \frac{\partial F^j}{\partial H^j}\frac{\partial H^j}{\partial p_2} & \dots & \frac{\partial F^j}{\partial H^j}\frac{\partial H^j}{\partial p_k} \end{bmatrix} \tag{11}$$

The last row in (9) was added to J^j_E so that consecutive Jacobian matrices of the sub-links can be multiplied together.

Except for the last sub-link, the lengths of the other sections do not change with respect to time. Only when the length of the last sub-link decreases to zero, the second sub-link towards the end starts to vary its length. Thus, except for the last sub-link, the Jacobian is not influenced by the linear motion of the concentric tube which results in $J^j_{g2} = 0$. This simplifies the corresponding Jacobian into

$$J^j_M = \begin{bmatrix} J^j_{g1} & 0 \\ 0 & I_{k\times k} \end{bmatrix} \tag{12}$$

For the last sub-link of the entire robot, the last row of the Jacobian matrix in (9) should be removed so that

$$J^j_L = \begin{bmatrix} J^j_{g1} & J^j_{g2} \end{bmatrix} \tag{13}$$

Using the chain rule, the Jacobian for the (m-1)th link $\tilde{J}^{m-1}$ is obtained by multiplication of individual Jacobians in the following manner.

$$\tilde{J}^{m-1} = J^n J^{n-1} ... J^2 J^1 \tag{14}$$

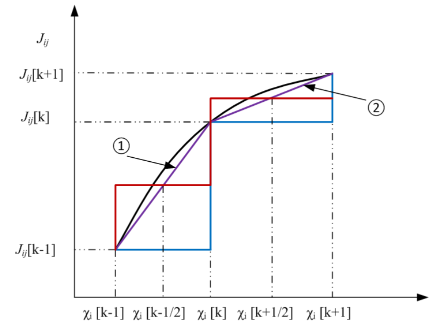


Fig. 3. Function approximation using a multi-step approach.

in which, J^j $(1 \leq j \leq n)$ is the Jocobian associated with the jth sub-link, and depending on the configuration, J^j is selected from the set $\{J^j_G, J^j_M, J^j_L\}$.

Thus, the relationship between the velocities of the distal and proximal ends is established as

$$[\dot{u}_{iz}(L)\ \dot{\Theta}(L)\ \dot{X}(L)]^T = \tilde{J}[\dot{u}_{iz}(0)\ \dot{\Theta}(0)\ \dot{X}(0)]^T \tag{15}$$

where $\tilde{J} = \tilde{J}^m \tilde{J}^{m-1} ... \tilde{J}^1$, and L is the total length of the assembled robot. To control the tip's position, only $\Theta(L)$ and $X(L)$ are required. However, based on the assumption of a torsionally compliant model outlined in section II-A, $u_{iz}(L)$ equals zero in any position, which leads to $\dot{u}_{iz}(L) = 0$. This constraint has to be considered during position control. From this perspective, the Jacobian derived in this section is actually an augmented Jacobian with constraint on velocities in Cartesian space.

C. *The Multi-Step Jacobian Method*

In surgical robotics, maintaining smooth and safe motion is critical and must be guaranteed. When the inverse kinematics is implicit, such as in concentric tubes, the tracking accuracy can be improved by updating the inverse Jacobian at a high rate which in turn increases computational complexity. Thus, the following approximation is suggested to obtain an accurate approximation of the robot's inverse kinematics which can generate a smooth trajectory even at low control rates. Let us revisit the forward kinematics as following

$$\chi(L) = [u_{iz}(L)\ \Theta(L)\ X(L)]^T = \tilde{F}(\chi(0)) \tag{16}$$

Performing a Taylor series expansion on (16), and keeping only the 1st order term, (16) is converted to (18) which is employed in an inverse Jacobian-based control scheme.

$$\Delta\chi(L) = \overbrace{\begin{bmatrix} \frac{\partial \tilde{F}}{\partial \chi_1(0)} & \frac{\partial \tilde{F}}{\partial \chi_2(0)} & \dots & \frac{\partial \tilde{F}}{\partial \chi_i(0)} \end{bmatrix}}^{\tilde{J}} .\Delta\chi(0) + H.O.T \tag{17}$$

$$\Delta\chi(L) \cong \tilde{J}.\Delta\chi(0) \Rightarrow \Delta\chi(0) \cong J^\dagger.\Delta\chi(L) \tag{18}$$

$J^\dagger$ is the inverse or pseudo-inverse of the Jacobian matrix that is used as a first order approximation of inverse kinematics. Inclusion of higher order terms (HOTs) is avoided in our implementation.

To have a fast approximation of the Jacobian while not sacrificing accuracy, the following procedure is exploited. Let us assume that the function shown in black in Fig. 3 is an arbitrary element of the Jacobian matrix, e.g., $\tilde{J}_{ij}$, whose value changes with respect to χ_i. During each control step, function $\tilde{J}_{ij}$ can be estimated using a zero-order hold shown by the blue line. The area between the black and blue curves corresponds to the approximation error, and it decreases as a higher control rate is chosen. Using a first-order hold represented by the purple curve also provides a more accurate solution, but it is avoided here due to computational issues. Our suggested technique is to introduce a new zero-order hold function shown in red which covers almost the same area as the first-order hold does, but is easier to be evaluated in each control step. The value of the first-order hold evaluated at the middle point in each iteration is chosen as the value of this new hold function. Therefore,

$$\tilde{J}_{ij}[k+\frac{1}{2}] = \tilde{J}_{ij}[k] + \frac{\tilde{J}_{ij}[k+1]-\tilde{J}_{ij}[k]}{\chi_i[k+1]-\chi_i[k]} \times \frac{\chi_i[k+1]-\chi_i[k]}{2} \quad (19)$$

At each discrete control step denoted by k, $\tilde{J}_{ij}[k+1]$ is unknown; so to evaluate $\tilde{J}_{ij}[k+\frac{1}{2}]$ in (19), it is assumed that the changes in the Jacobian is almost the same during two successive iterations, i.e., $\tilde{J}_{ij}[k+1]-\tilde{J}_{ij}[k] \cong \tilde{J}_{ij}[k]-\tilde{J}_{ij}[k-1]$. Thus,

$$\tilde{J}_{ij}[k+\frac{1}{2}] \cong \frac{3}{2}\tilde{J}_{ij}[k] - \frac{1}{2}\tilde{J}_{ij}[k-1] \quad (20)$$

At each time instant, the following approximation holds for the elements of the Jacobian matrix which ultimately results in a multi-step Jacobian position control scheme. Due to the complexity of the $\tilde{J}$ elements, all calculations were performed using Maple® software.

$$\Delta\chi(0) \cong \left(\frac{3}{2}\tilde{J}[k] - \frac{1}{2}\tilde{J}[k-1]\right)^{\dagger}.\Delta\chi(L) \quad (21)$$

III. SIMULATION STUDY

Before the results are presented, it is worth noting that when updating the Jacobian, the boundary condition acts as an input signal, most of which is known except for $u_{iz}(0)$. One practical way to resolve this problem is to measure this value using a force/torque sensor mounted at the robot's base; however, this approach is not applicable for a simulation study. Fig. 4 illustrates the trend of $\beta = u_{iz}(L)$ with respect to $\eta = u_{iz}(0)$ for a given rotation angle α between the tubes in a two-tube robot. In six simulated configurations, the mapping turns out to be monotonic and roughly linear. Therefore, a piecewise linear approximation as in (22) is adopted to find the slope of this mapping. Using (23), $u_{iz}(0)$ can be obtained iteratively starting from an arbitrary value of $u_{iz}(L)$. Here, k represents the iteration number.

$$\begin{bmatrix} u_{iz}(s) \\ \alpha_i(s) \end{bmatrix} = G(u_{iz}(0), \alpha_i(0), s)$$
$$\Rightarrow \begin{bmatrix} \dot{u}_{iz}(L) \\ \dot{\alpha}_i(L) \end{bmatrix} = \begin{bmatrix} \frac{\partial G_{1..n}}{\partial u_{iz}} & \frac{\partial G_{1..n}}{\partial \alpha_i} \end{bmatrix} \cdot \begin{bmatrix} \dot{u}_{iz}(0) \\ \dot{\alpha}_i(0) \end{bmatrix} \quad (22)$$

$$\eta[k+1] = \eta[k] - \left(\frac{\partial G_{1..n}}{\partial u_{iz}}\right)^{-1}.(\beta[k]-0) \quad (23)$$

The diameters, lengths, curvatures, and mechanical properties of the three tubes used for the simulation of a three-tube robot are listed in Table I in which the lengths of straight and curved sections of the tubes are represented by l and L, respectively. The radius of curvature is also denoted by r. We also simulated a two-tube robot consisting of the middle and outer tubes defined in Table I. These parameters are the actual parameters of our two-tube concentric robot that is experimented in section IV.

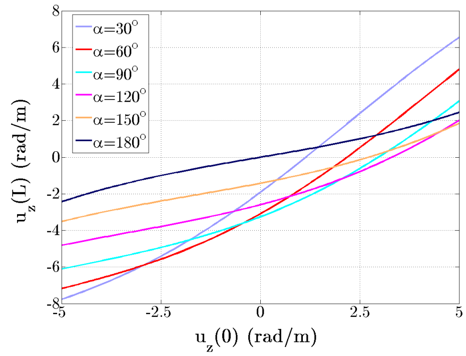


Fig. 4. The relationship between boundary conditions: $u_{iz}(L)$ with respect to $u_{iz}(0)$ as a function of the rotation angle in a two-tube robot.

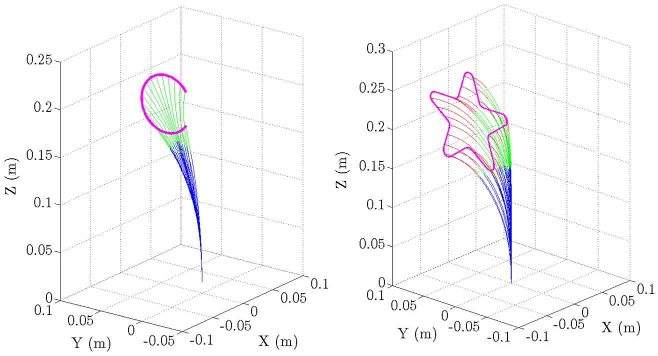
TABLE I
PARAMETERS OF THE TUBES USED FOR THE SIMULATION

Tube pair	1 (inner)	2 (middle)	3 (outer)
l (mm)	250	155	12
L (mm)	150	152.6	152.6
r (mm)	100	250	250
Stiffness ratio (compared to the 3rd tube)	0.25	0.85	1

In the simulation study for the three-tube robot, $u_{iz}(0)$ converged successfully within three iterations while the calculation time was negligible. This method was initially developed for a two-tube robot, and it can be extended to any number of tubes. At the next step, the Jacobian-based control scheme was evaluated by running a trajectory tracking test whose results are shown in Fig. 5. The desired trajectory vector in this simulation was defined by $[u_{iz}(L) \equiv 0 \;\; \Theta_{tr}(L) \;\; X_{tr}(L)]^T$, and the two-tube and three-tube robots tracked the designed trajectory accurately. In the graphs each sub-link's length was set to be 10mm.

We ran the tracking simulation 1000 times. It took total of 1.3sec to calculate the Jacobian and the forward kinematics in the two-tube robot case using 10-mm sub-links. The same calculation for the three-tube robot took less than 3sec on a desktop computer with a dual-core 2.4GHz 32 Bit processor.

To compare the improvement in trajectory tracking using the introduced multi-step Jacobian technique, the two-tube robot was simulated to follow a straight line whose length was 80mm. First, each Jacobian update size was set to 1mm. The modified Jacobian reduced the mean tip positioning error from 2.15×10^{-2}mm to 1.14×10^{-3}mm compared to the conventional Jacobian matrix. Setting the update step to be 8mm which corresponds to a low update rate in a control system, the mean tracking error was reduced from approximately 1.5mm to 0.6mm. Corresponding results are shown in Fig. 6.



(a) (b)

Fig. 5. Trajectory tracking using (a) two-tube (blue:outer tube, green: middle tube) (b) three-tube concentric robots (blue:outer tube, green: middle tube, red: inner tube).

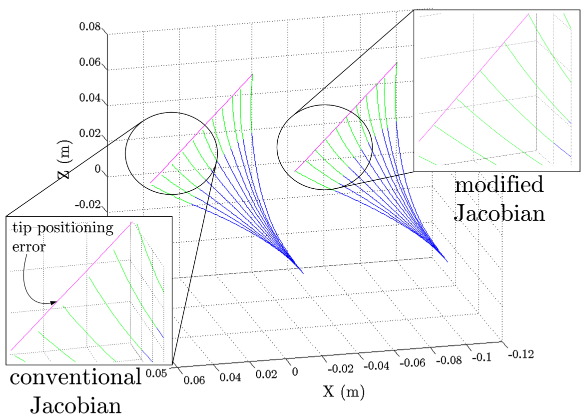


Fig. 6. Trajectory tracking using conventional Jacobian and modified multi-step Jacobian approach in the two-tube robot.

IV. EXPERIMENTAL VALIDATION

A. Setup Description

The experimental evaluation of the proposed Jacobian-based control approach was carried out using the concentric-tube robot composed of two superelastic tubes (see Fig. 7). The inset drawing in Fig. 7 illustrates the Nitinol tubes each of which has a straight section at its proximal end followed by a distal section with a constant curvature.

The outer tube was rotated with a rotary T-RS60 stage while the inner tube was respectively inserted and rotated using a linear T-LSR300B stage and another rotary T-RS60 stage (Zaber). The robot was equipped with a Nano43 6-DOF force/torque sensor (ATI Industrial Automation) located at the base of the inner tube. This sensor was used to align the tubes knowing the fact that the the torsion is minimized when the tubes are fully aligned. The angle difference between the two tubes was obtained in this way as 306.3°. In the setup, an Aurora EM (electromagnetic) tracking system (Northern Digital Inc.) was used for determining position during tracking; however, the control algorithm was updated using the tip's position provided by the forward kinematics. This strategy gave more stable results. A sensor coil of the EM tracker was attached to the tip of the inner tube by a light plastic adaptor. A multi-threaded application for real-time control was developed using Microsoft® C++, MATLAB®

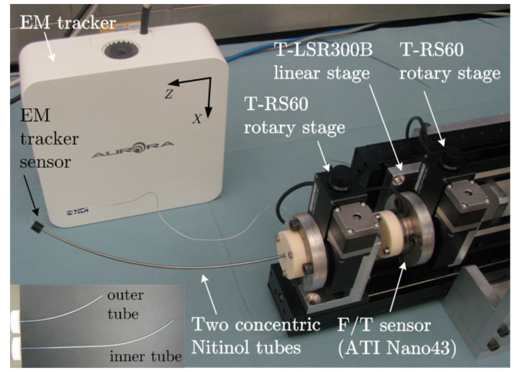


Fig. 7. Our concentric-tube robot setup.

and the QuaRC® Toolbox (Quanser Inc.).

We utilized the modified Jacobian approach with damped least-squares method as a classical approach to avoid singularities during trajectory tracking [11]. However, the main disadvantage of this singularity-avoidance technique is an increase in positioning error. A detailed analysis of the singularity and manipulability measures for concentric-tube robots is part of our ongoing research.

B. Experiments and Results

We now provide some experimental results using the scheme described above for our concentric-tube robot with two tubes. In our test-bed, three experiments were conducted to evaluate trajectory tracking. In these cases, the initial rotary angles are set at $[0\ 60^{o}]$ which are very close to a singularity of the robot. This enabled us to test the singularity-avoidance scheme that was implemented since we encountered several singular configurations.

In the first experiment, a straight line in the XY plane of the setup's coordinate frame indicated in Fig. 7 was required to be followed by the robot's tip in 40 seconds. The tip is also expected to remain stationary in the XZ and YZ planes. Fig. 8 displays the results. Accordingly, the root mean squared error in the entire 3D plane and the final positioning error at the tip's location were 1.79mm and 3.86mm, respectively. As seen in Fig. 8, singularity avoidance was maintained which generated a small curved path in the vicinity of the origin in the XY plane subplot. As the next test, a sinusoidal motion profile in the XY plane was used as the reference trajectory. The desired and experimental paths followed by the robot's tip are presented in Fig. 9. The error values in this case were 1.56mm and 1.67mm, respectively. In both cases, the length of the sub-links was chosen to be 10mm. In the final experiment, a combination of linear and sinusoidal trajectories was utilized to develop a desired 3D trajectory. The trajectory resulting from the experiment indicates good tracking performance (see Fig. 10).

The results obtained confirm the capability of the suggested scheme to enable the robots tip to track desired trajectories with an acceptable accuracy. However, there are deviations from the desired paths which can be attributed to (1) measurement error of the EM tracker which has been reported to be around 1mm, (2) the forward kinematics error

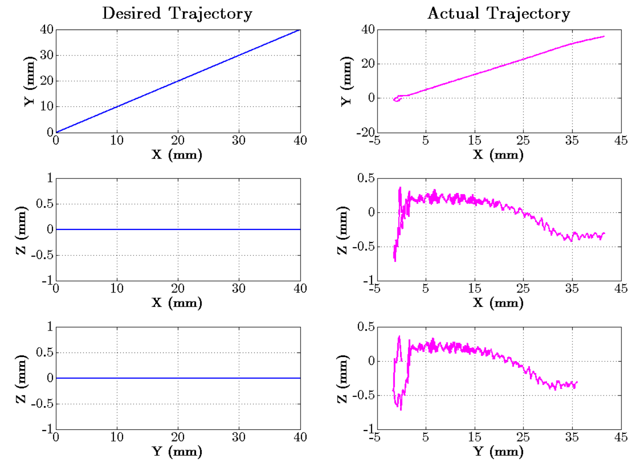


Fig. 8. Desired and experimental trajectories in XY, XZ, and YZ planes (linear trajectory, final tip position = $[41.59\ \ 37.51\ \ -0.31]^T$ mm).

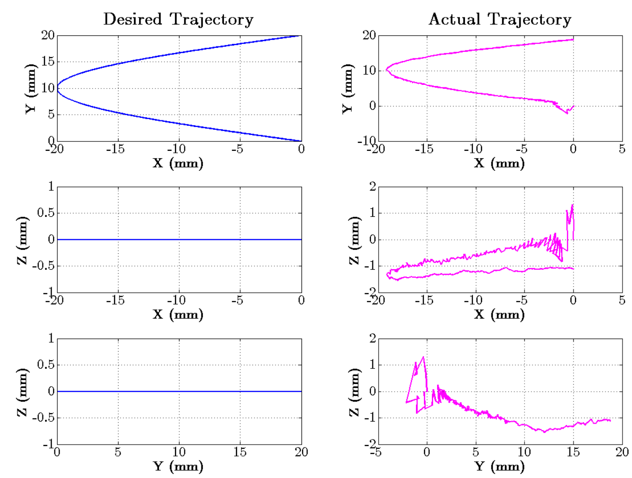


Fig. 9. Desired and experimental trajectories in XY, XZ, and YZ planes (sinusoidal motion, final tip position = $[-0.02\ \ 18.76\ \ -1.12]^T$ mm).

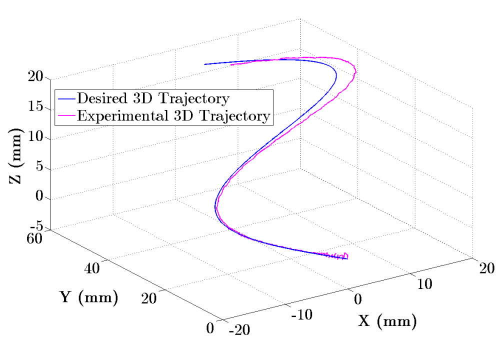


Fig. 10. Desired and experimental trajectories in 3D (final tip position = $[2.329\ \ 47.79\ \ 19.65]^T$ mm, final desired position = $[0\ \ 50\ \ 20]^T$ mm).

which is estimated to have an average value of 1.5mm using similar mechanical parameters in a two-tube robot [10]; and (3) inclusion of the singularity-avoidance technique. Furthermore, ignoring unmodeled dynamics, such as frictional torques [12], as have done in this study, may also have contributed to the positioning and tracking inaccuracies.

V. CONCLUSION AND FUTURE WORK

A concentric-tube robot is well-suited for navigation along 3D complex curves since the robot's shape is sufficiently flexible. Using computer-assisted techniques, the shape of this device can be accurately controlled to guide it inside the natural orifices, lumens, and other anatomical areas in a variety of minimally invasive applications. From the control perspective, real-time positioning of this type of robots is challenging. However, addressing some of the computational issues makes it possible to develop a variety of closed-loop control strategies that can significantly increase the operational accuracy and performance of these robots.

The control approach presented in this paper can be implemented efficiently. The proposed Jacobian-based controller was shown to provide reasonable performance without the need for excessive online or pre-computation. In many clinical applications such as biopsies, a 2- to 4-mm positioning error is tolerable, whereas in sensitive applications, e.g., retinal microsurgery, a higher level of accuracy is required. In our implementation, friction compensation was not considered. However, predicting frictional torques as described in [12] is expected to improve the overall tracking performance. That is left for future work. Another enhancement for a future study is to incorporate the effect of an external load.

VI. ACKNOWLEDGMENTS

The authors thank I. Khalaji and C. Vandelaar for their assistance in fabricating the concentric-tube robot.

REFERENCES

[1] A. Asadian, M.R. Kermani, and R.V. Patel, "Robot-assisted Needle Steering using a Control Theoretic Approach," *J. Intell. Robot. Syst.*, vol. 62, no. 3-4, pp. 397-418, 2011.

[2] P. Dupont, J. Lock, B. Itkowitz, and E. Butler, "Design and Control of Concentric-tube Robots," *IEEE Trans. Robot.*, vol. 26, no. 2, pp. 209-225, 2010.

[3] R.J. Webster III, J.M. Romano, and N.J. Cowan, "Mechanics of Precurved-tube Continuum Robots," *IEEE Trans. Robot.*, vol. 25, no. 1, pp. 67-78, 2009.

[4] P. Sears and P.E. Dupont, "A Steerable Needle Technology using Curved Concentric Tubes," *In Proc. of IEEE/RSJ Int. Conf. on Intel. Rob. Sys. (IROS)*, 2006, pp. 2850-2856.

[5] P. Sears and P.E. Dupont, "Inverse Kinematics of Concentric Tube Steerable Needles," *In Proc. of IEEE Int. Conf. on Rob. Autom. (ICRA)*, 2007, pp. 1887-1892.

[6] R.J. Webster III, J.P. Swensen, J.M. Romano, and N.J. Cowan, "Closed-form Differential Kinematics for Concentric-tube Continuum Robots with Application to Visual Servoing," *In Proc. of 11th Int. Symp. on Experimental Rob.*, 2008, *Springer Tracts in Advanced Robotics*, 2009, 54, pp. 485-494.

[7] D.C. Rucker and R.J. Webster III, "Mechanics-based Modeling of Bending and Torsion in Active Cannulas," *In Proc. of IEEE RAS/EMBS Int. Conf. on Biomed. Rob. Biomech. (BioRob)*, 2007, pp.704-709.

[8] J. Lock, G. Laing, M. Mahvash, and P.E. Dupont, "Quasistatic Modeling of Concentric Tube Robots with External Loads," *In Proc. of IEEE/RSJ Int. Conf. on Intel. Rob. Sys. (IROS)*, 2010, pp. 2325-2332.

[9] D.C. Rucker, B.A. Jones, and R.J. Webster III, "A Geometrically Exact Model for Externally Loaded Concentric-tube Continuum Robots," IEEE Trans. Robot., vol. 26, no. 5, pp. 769-780, 2010.

[10] R. Xu and R.V. Patel, "A Fast Torsionally Compliant Kinematic Model of Concentric-tube Robots," *In Proc. of 34th IEEE EMBS Annu. Int. Conf.*, 2012, pp. 904-907.

[11] G. Marani, *et. al*, "A Real-time Approach for Singularity Avoidance in Resolved Motion Rate Control of Robotic Manipulators," *In Proc. of IEEE Int. Conf. on Rob. Autom. (ICRA)*, 2002, pp. 1973-1978.

[12] J. Lock and P.E. Dupont, "Friction Modeling in Concentric Tube Robots," *In Proc. of IEEE Int. Conf. on Rob. Autom. (ICRA)*, 2011, pp. 1139-1146.